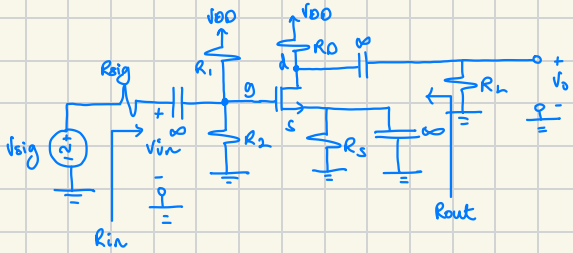
ELG 2136

lecture 8

AC analysis of MOSFET



- disable DC sources
- short circuit all capacitors
- Replace MOSFET by small-signal model.

$A_v = v_o/v_i$?

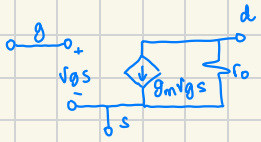
R_{in} ?

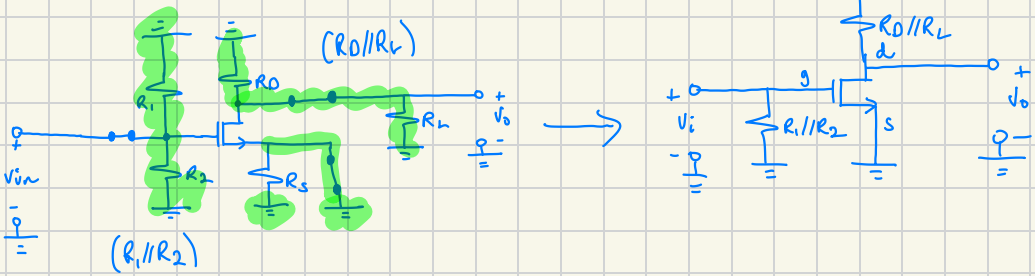
$G_v = v_o/v_{sig}$?

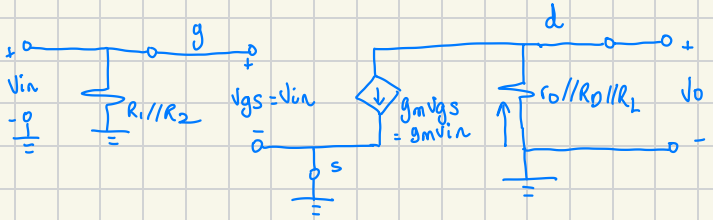
$A_i = i_o/i_i$?

R_{out} ?

small signal model:

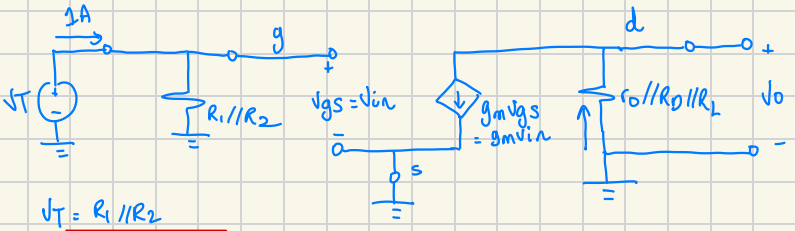






$$g_m v_{in} = \frac{0 - v_o}{r_o // R_D // R_L} \Rightarrow g_m v_{in} = \frac{-v_o}{r_o // R_D // R_L}$$

$$\Rightarrow \boxed{\frac{v_o}{v_i} = A_v = -g_m [r_o // R_D // R_L]}$$

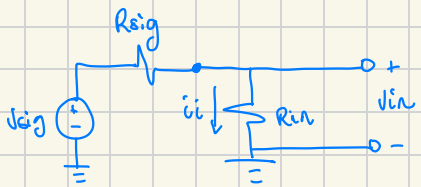


$V_T = R_1 // R_2$

$\Rightarrow$ **$R_{in} = R_1 // R_2$**

$$G_V = \frac{V_o}{V_{sig}} = \left(\frac{V_o}{V_{in}}\right)\left(\frac{V_i}{V_{sig}}\right) = A_V \left[\frac{V_i}{V_{sig}}\right]$$

($\frac{V_o}{V_{in}} = A_V$)



$$V_{in} = \left(\frac{R_{in}}{R_{sig}+R_{in}}\right) V_{sig}$$

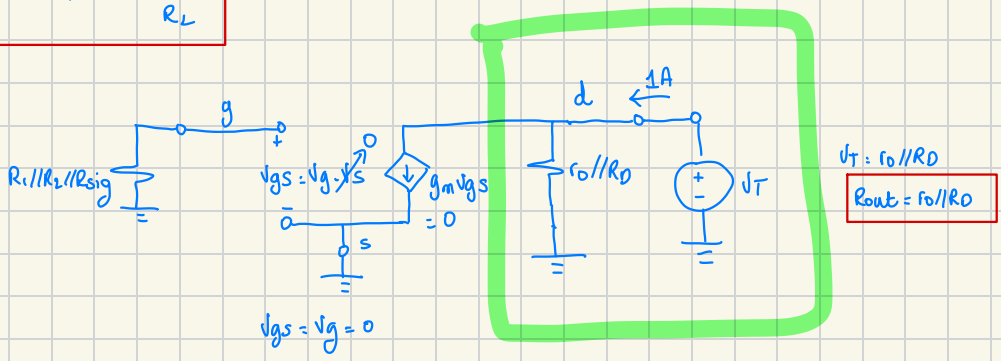
$$\Rightarrow \frac{V_{in}}{V_{sig}} = \frac{R_{in}}{R_{sig}+R_{in}}$$

$$\boxed{G_V = -g_m [r_o // R_D // R_L] \left(\frac{R_1 // R_2}{R_{sig} + R_1 // R_2}\right)}$$

$$A_i = \frac{i_o}{i_i}, \quad i_i = \frac{v_i}{R_{in}}, \quad i_o = \frac{v_o}{R_L}$$

$$A_i = \left(\frac{v_o}{R_L}\right)\left(\frac{R_{in}}{v_i}\right) = \left(\frac{R_{in}}{R_L}\right) A_v = -g_m [r_o // R_D // R_L] \frac{R_1 // R_2}{R_L}$$

$$\boxed{A_i = -g_m [r_o // R_D // R_L] \frac{R_1 // R_2}{R_L}}$$



$V_T = r_o // R_D$

$R_{out} = r_o // R_D$